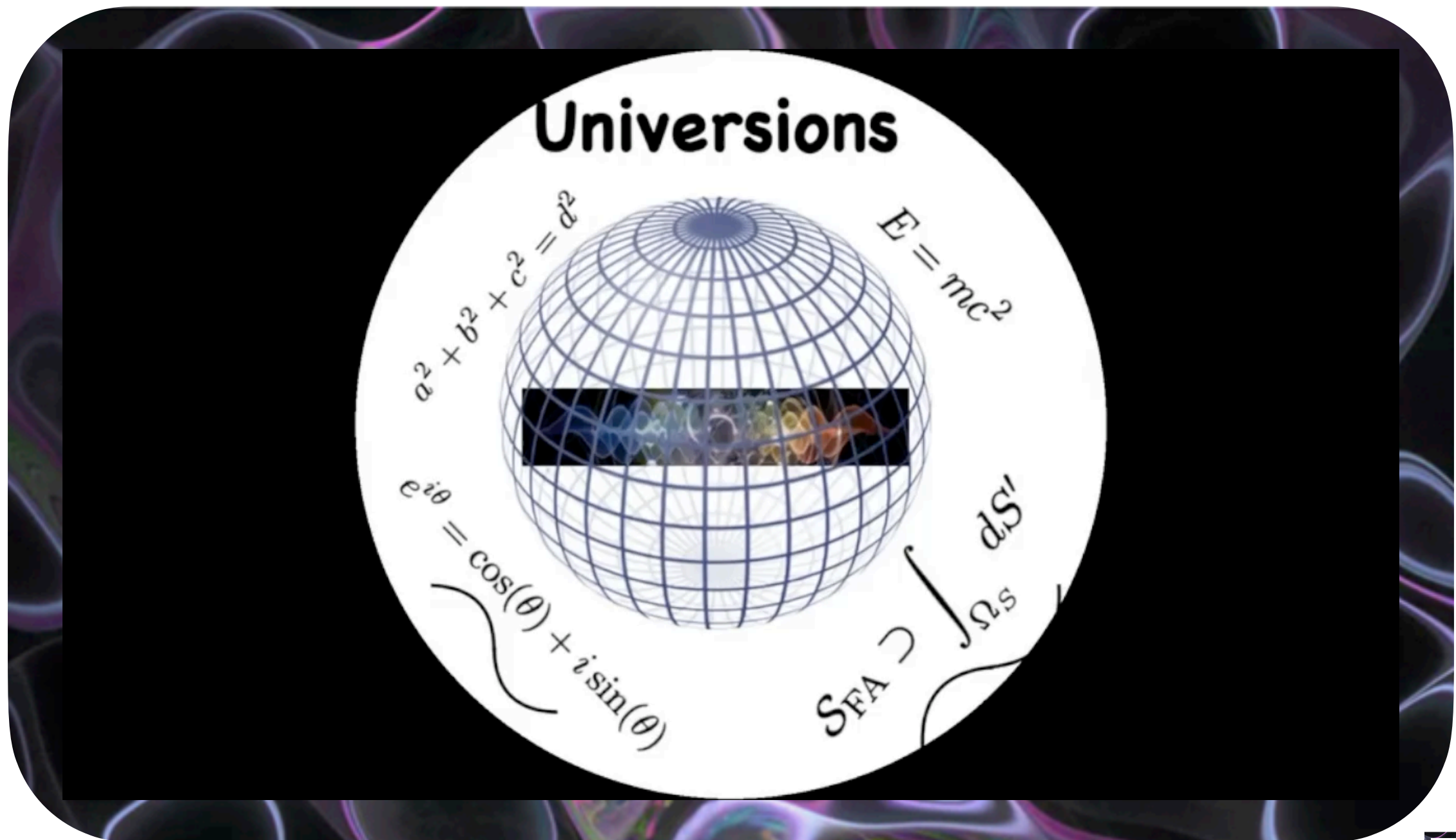


Analytical poly Λ CDM in cosmic observations



Outline

Introduction

Theoretical framework

- poly Λ CDM (in cosmic observations)
- Functors of Actions Theories (FAT) (Cosmos APP)
- Quintessence
- A temporal-spatial-entropic K curvature ([EinsteinKPy2Tex code](#))
- Extra Dimensions in Observations
- Advanced Manifold-Metrics Pairs (AMMP)
- Advanced Tensor Theories (ATT)
- A Foundational Analysis of Kernel-based Calculus (AFAKEC)
- A cosmic femtolensing Hawking radiation (Poster)

Recent
results

Conclusions and future

COsmological GRAvitological TheORy (COGRATORY)

Ezquiaga & Zumalacárregui 18
and novel theories

Functors
of
Action
Theories
 S_{FAT}

**General
Relativity**

unique theory of
massless $g_{\mu\nu}$

**Massive
Gravity**
 $m_g > 0$

Tensor
 $T_{\mu\nu}$

$$S = \int_{\mathcal{V}} d^D x \sqrt{-g} \mathcal{L}$$

**Additional
Field**

Vector V_μ

Scalar
 ϕ

Love-
lock

Quint-
essence

Brans-
Dicke

Hordenski

$f(R)$

Gauss-
Bonnet

Galileon
UEDM

Galileon

KGB

**Beyond
Hordenski**

$C(X)$

$D(X)$

$E(X)$

**Break
Assumptions**

Mathly
interesting

Advanced
Manifold-Metric
Pairs

Fractional
Calculus

Advanced
Tensor
Theories

Entropic
Dimensions

Extra
Dimensions

Probabilistic
Dimensions

Strings
/Branes

Backreaction

(Non)Isotropy
(Non)homogeneity

(Non)locality

(Non)Causality

Non-Riemanniann

Curvature
-types
 $f(R^n)$

Torsion
 $f(T)$

non-
metricity
 $f(Q)$

Combined
 $f(R,T,Q)$

COsmological GRAvitological TheORy (COGRATORY)

Ezquiaga & Zumalacárregui 18
and novel theories

Functors
of
Action
Theories
 S_{FAT}

General
Relativity

unique theory of
massless $g_{\mu\nu}$

Massive
Gravity
 $m_g > 0$

Tensor
 $T_{\mu\nu}$

Additional
Field

Vector V_μ

Scalar
 ϕ

Love-
lock

Quint-
essence

Brans-
Dicke

Hordenski

$f(R)$

Gauss-
Bonnet

Galileon
UEDM

Galileon

KGB

Beyond
Hordenski

$C(X)$

$D(X)$

$E(X)$

Break
Assumptions

Advanced
Manifold-Metric
Pairs

Mathly
interesting

Advanced
Tensor
Theories

Fractional
Calculus

Entropic
Dimensions

Probabilistic
Dimensions

Extra
Dimensions

Strings
/Branes

Backreaction

(Non)Isotropy
(Non)homogeneity

(Non)locality

(Non)Causality

Non-Riemanniann

Curvature
-types
 $f(R^n)$

Torsion
 $f(T)$

non-
metricity
 $f(Q)$

Combined
 $f(R,T,Q)$

$$S = \int_V d^D x \sqrt{-g} \mathcal{L}$$

Outline

Introduction

Theoretical framework

- poly Λ CDM (in cosmic observations)

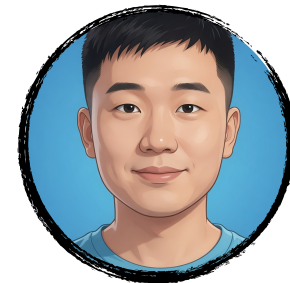
Recent results

- Functors of Actions Theories (FAT) (Cosmos APP)
 - Quintessence
 - Extra Dimensions in Observations
 - Advanced Manifold-Metrics Pairs (AMMP)
 - A temporal-spatial-entropic K curvature ([EinsteinKPy2Tex code](#))
 - Advanced Tensor Theories (ATT)
 - A Foundational Analysis of Kernel-based Calculus (AFAKEC)
-
- A cosmic femtolensing Hawking radiation (Poster)

Conclusions and future



Which effective model can help us distinguish among Modified Gravity Cosmological Models ?



ZX & PN
under
prep.



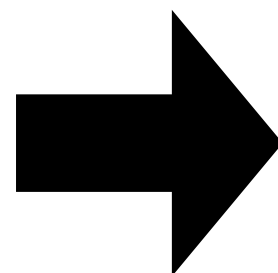
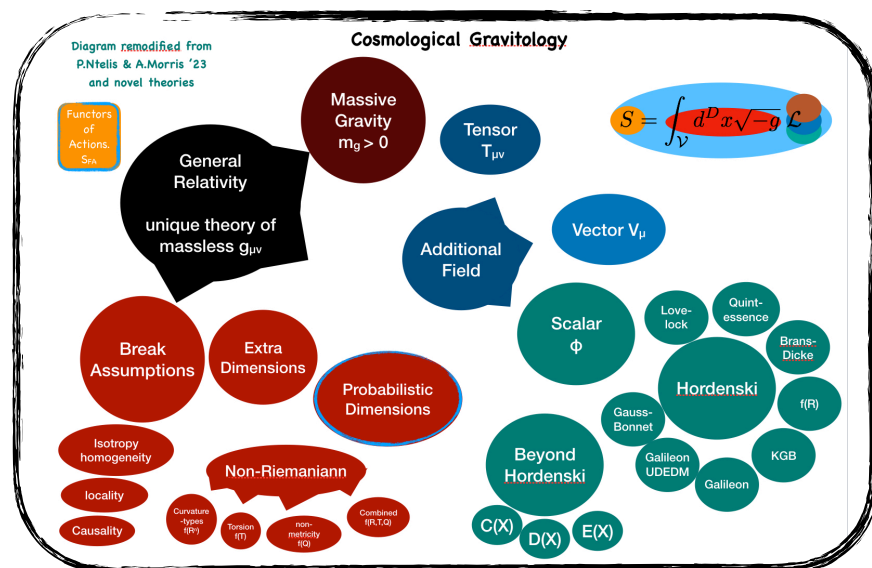
Analytical *poly* Λ CDM dynamics

Pierros Ntelis^a, Jackson Levi Said^{b,c}

^aIndependent Research Affiliation Former Aix Marseille Univ. Marseille France

^bInstitute of Space Sciences and Astronomy University of Malta Msida Malta

^cDepartment of Physics University of Malta Msida Malta



Standard
 Λ CDM

+

Modified
Gravity
and DE

Or in details

$$L = (m, r, \Lambda) + (IDE, f(R), \text{Curvature}, SVT)$$

... Further Simulations ...

$$H(a) = \sum_{s \in \{m, r, \Lambda, x, k, z, v\}} \Omega_{s,0} a^{-\alpha_s}$$

Analytical poly Λ CDM dynamics

From Action to Effective Lagrangians which model Modified Gravity models

$$H(z) = H_0 \left[\begin{array}{ll} \Omega_{m,0}(1+z)^3 & \text{Matter} \\ +\Omega_{r,0}(1+z)^4 & \text{Radiation} \\ +\Omega_{x,0}(1+z) & \text{f(R)} \\ +\Omega_{k,0}(1+z)^2 & \text{Curvature} \\ +\Omega_{z,0}(1+z)^5 & \text{Interacting Dark Energy} \\ +\Omega_{v,0}(1+z)^\alpha & \text{SVT} \\ +\Omega_{\Lambda,0} & \text{Cosmo Constant} \end{array} \right]^{1/2}$$

Analytical poly Λ CDM dynamics

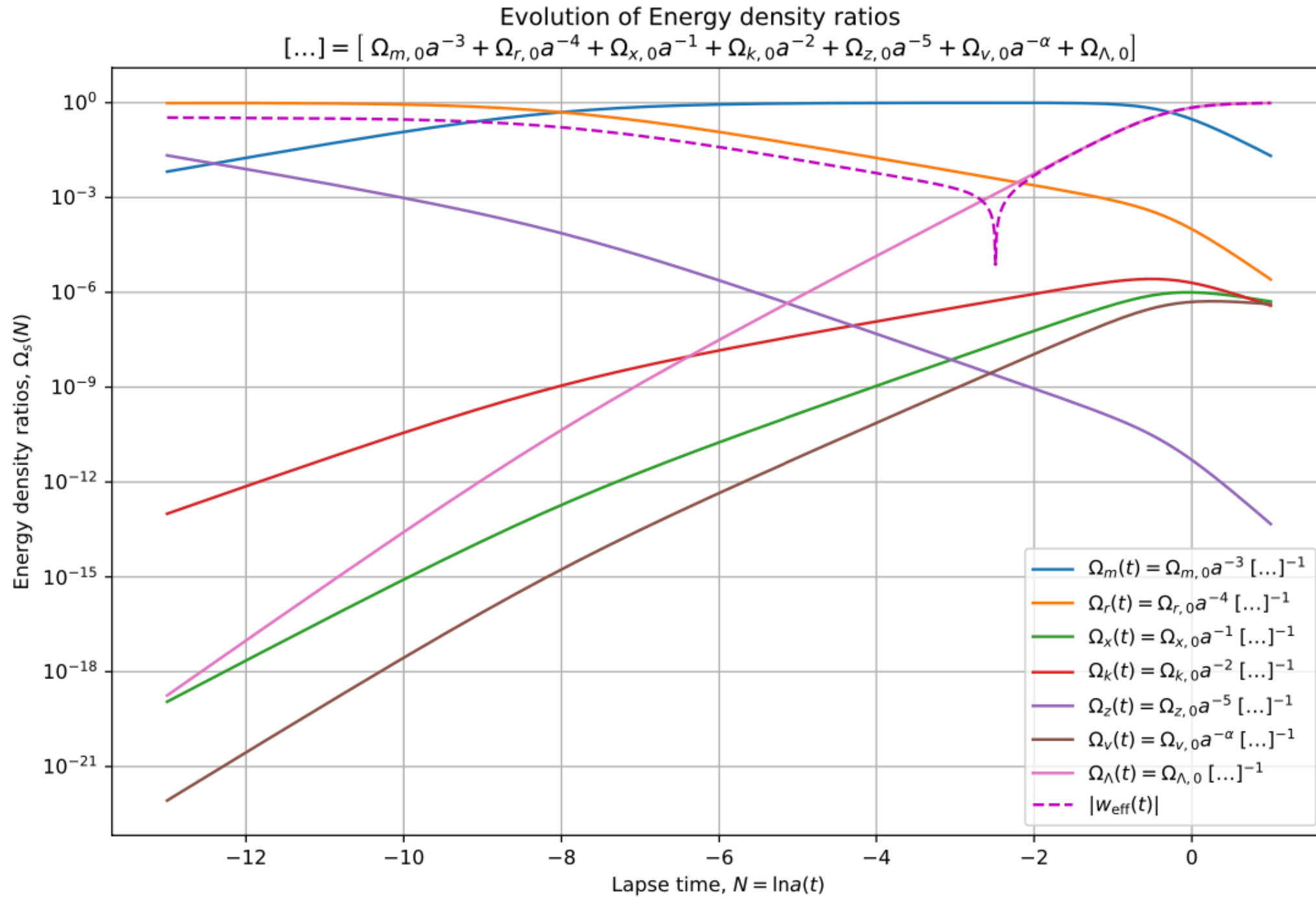


Figure 5: Energy density ratio of different species, Ω_s versus lapse time, N , of *poly* Λ CDM.
[See section [3.3.2](#)]

Analytical $\text{poly}\Lambda\text{CDM}$ dynamics

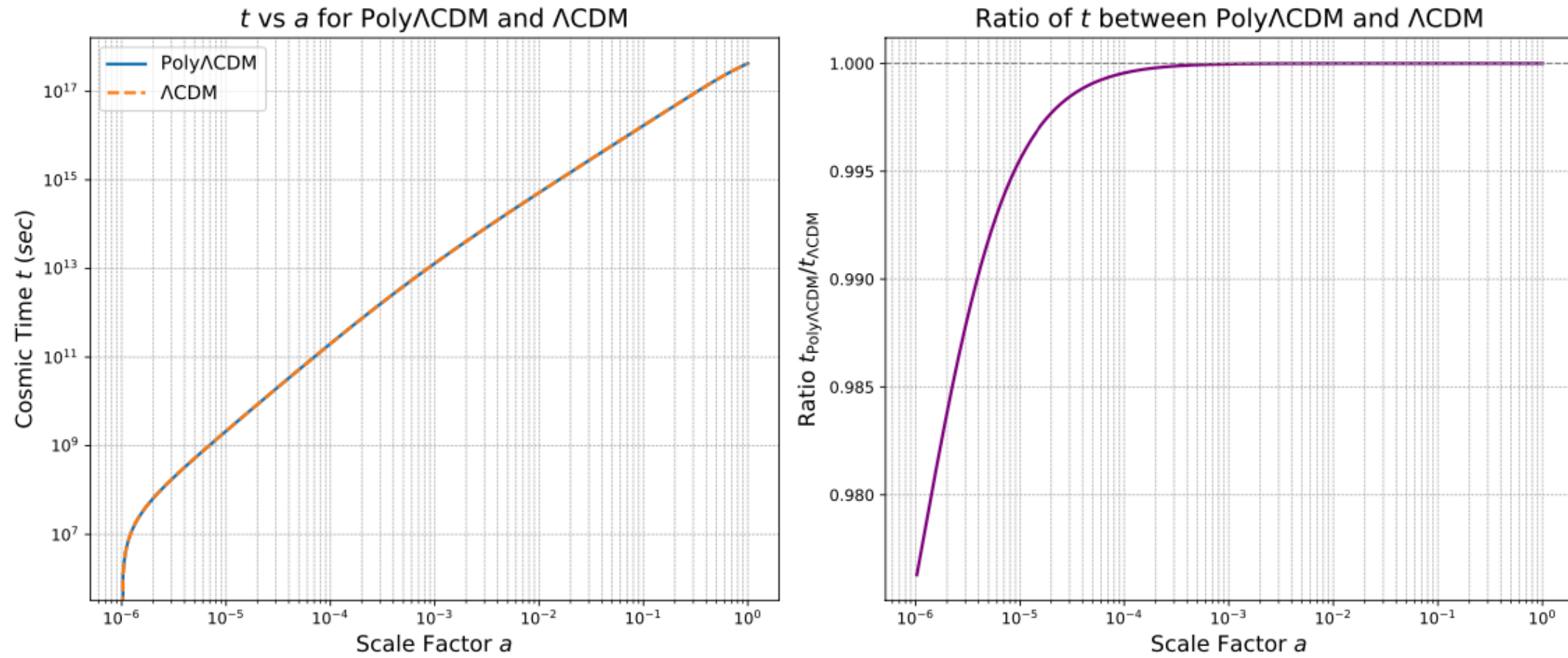


Figure 6: Left: Time, t versus scale factor, a , for the 2nd Friedman equation, of $\text{poly}\Lambda\text{CDM}$ and of ΛCDM . Right: Ratio of time, t , between $\text{poly}\Lambda\text{CDM}$ and ΛCDM versus scale factor, a . [See section 3.3.3]

for $\text{poly}\Lambda\text{CDM}$ cosmology is

$$t - t_0 = H_0^{-1} \int_{a_0}^a \frac{da}{a [\sum_s \Omega_{s,0} a^{-\alpha_s}]^{\frac{1}{2}}}$$

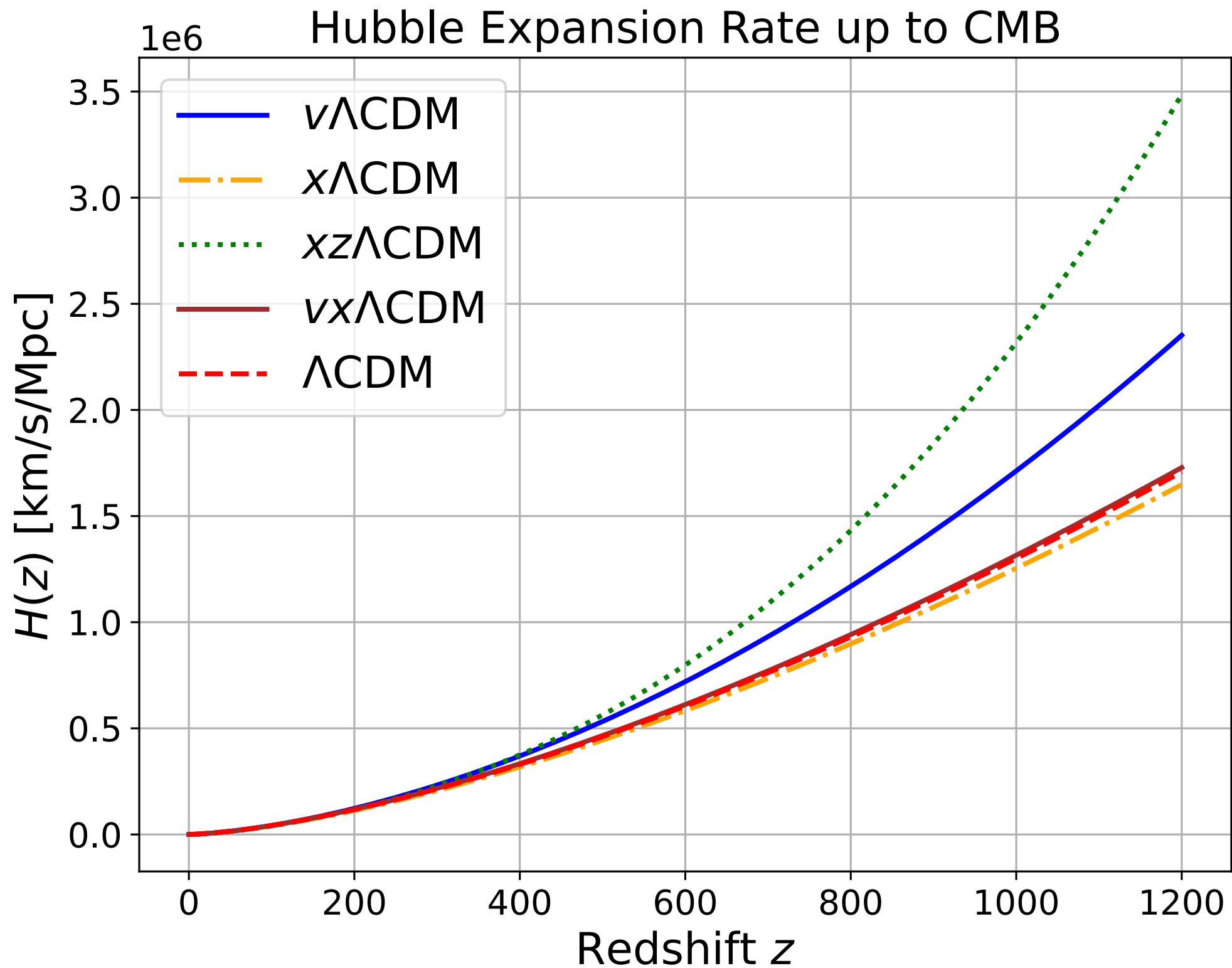
where $a_0 = 10^{-6}$.

ΛCDM cosmology, which is

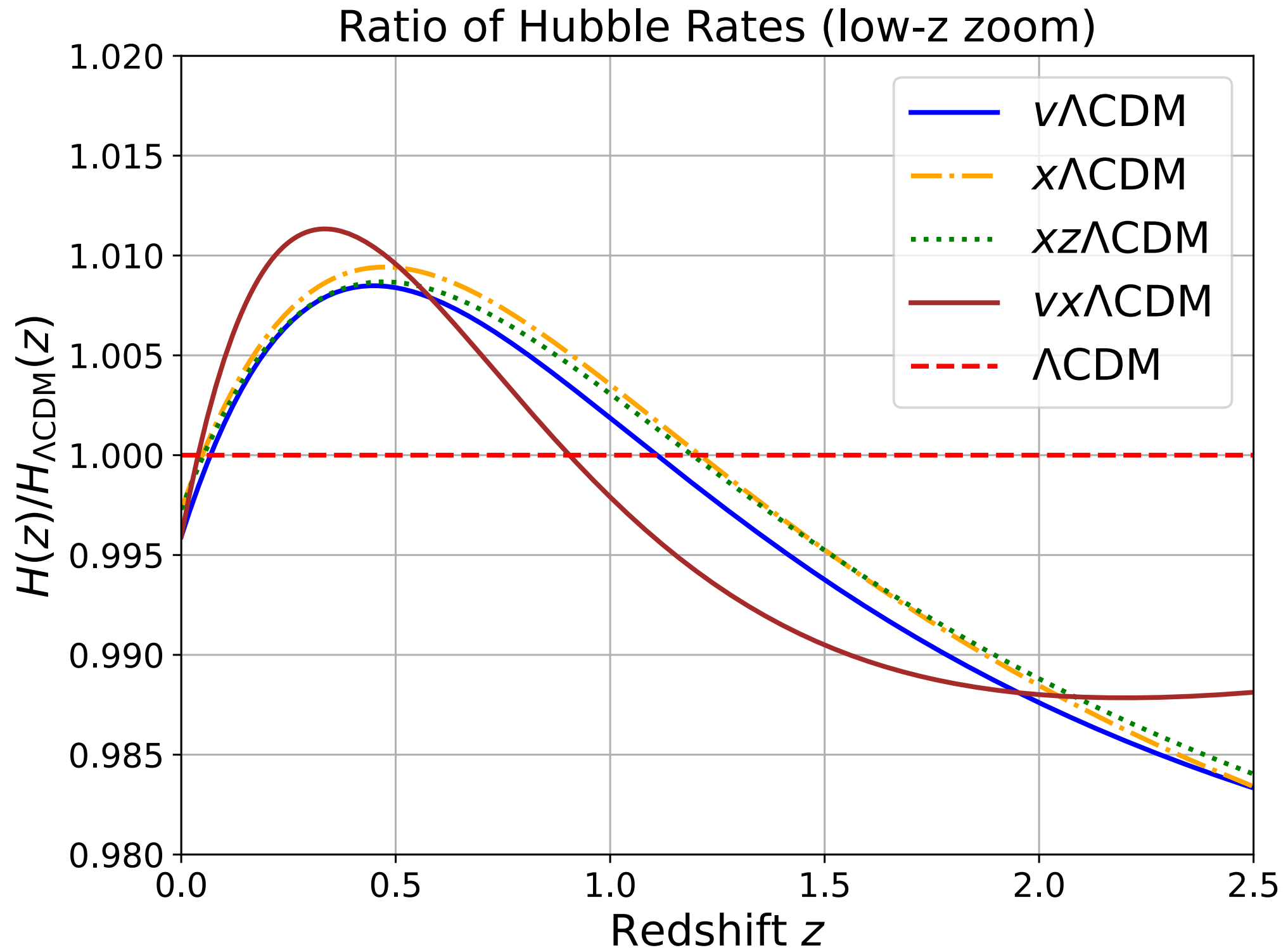
$$t - t_0 = H_0^{-1} \int_{a_0}^a \frac{da}{a [\Omega_{m,0} a^{-3} + \Omega_{r,0} a^{-4} + \Omega_{\Lambda,0}]^{\frac{1}{2}}} \quad (67)$$

Potential Observational Constrains from $H(z)$: BAO: $d_A(z)$, $d_H(z)$, and SNe: $d_L(z)$

Analytical poly Λ CDM dynamics



Analytical poly Λ CDM dynamics



Analytical poly Λ CDM dynamics

$$\chi_{\text{total}}^2 = \chi_{\text{SNIa}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{CMB}}^2$$

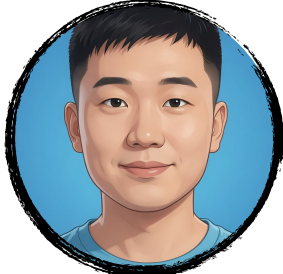
TABLE III. Model comparison statistics. ΔAIC and ΔAICc are defined as ΛCDM minus the alternative model, so positive values favour the alternative. The guidelines for interpreting $|\Delta|$ are given in Appendix A.

Model	χ_{min}^2	k	AIC	AICc	ΔAIC	ΔAICc
ΛCDM	1538.53	5	1548.53	1548.60	0.00	0.00
$v\Lambda\text{CDM}$	1535.49	6	1547.49	1547.53	1.04	1.07
$x\Lambda\text{CDM}$	1535.66	6	1547.66	1547.70	0.87	0.90
$xz\Lambda\text{CDM}$	1535.68	7	1549.68	1549.73	-1.15	-1.13
$vx\Lambda\text{CDM}$	1534.66	7	1548.66	1548.71	-0.13	-0.11

 Better performance

 Worse performance

ΛCDM still remains, but **Update** DATA/Method ...

  ZX & PN
under
prep.

Outline

Introduction

Theoretical framework

- poly Λ CDM (in cosmic observations)

Recent
results

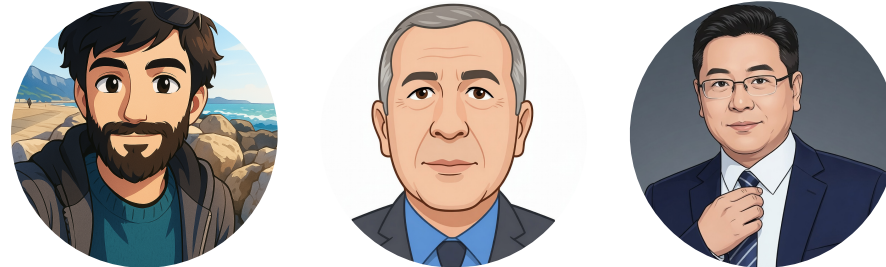
- Functors of Actions Theories (FAT) (Cosmos APP)
- Quintessence
- Extra Dimensions in Observations
- Advanced Manifold-Metrics Pairs (AMMP)
- A temporal-spatial-entropic K curvature ([EinsteinKPy2Tex code](#))
- Advanced Tensor Theories (ATT)
- A Foundational Analysis of Kernel-based Calculus (AFAKEC)

- A cosmic femtolensing Hawking radiation (Poster)

Conclusions and future

A cosmic femtolensing Hawking radiation (Poster)

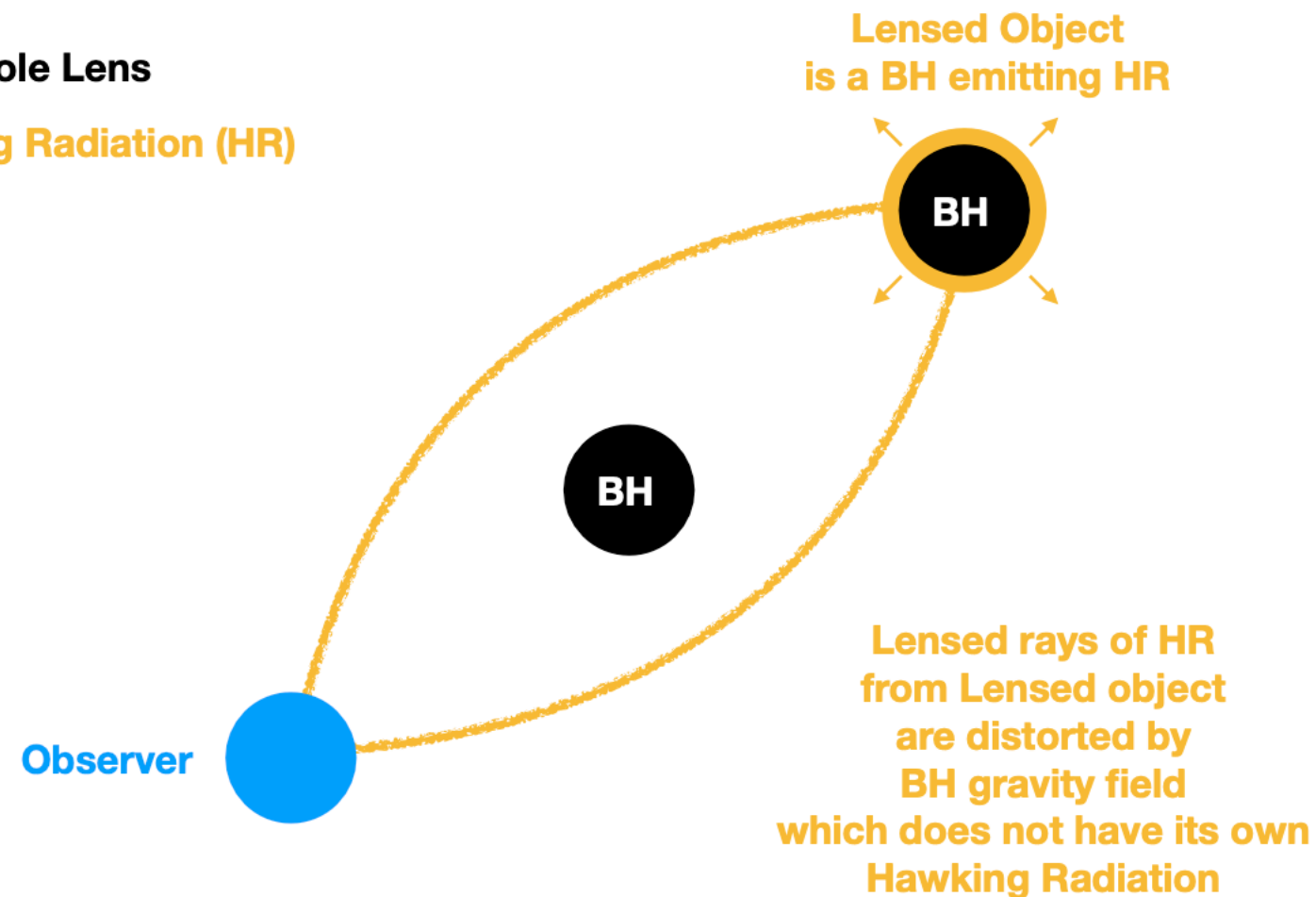
PN, B.J. Ahmedov, C. Yuan



Light Path: Lensed Object with HR -> Black Hole Lens without HR -> Observer

Black Hole Lens

Hawking Radiation (HR)



Thank you

谢谢

Xiè xiè

Ευχαριστώ

! :))

Functors of actions

Pierros Ntelis^{1, a)} and Adam Morris^{2, b)}

¹⁾ *Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France*

²⁾ *Universität Bonn - Helmholtz-Institut für Strahlen und Kernphysik, Bonn, Germany*

(Dated: 24 February 2023)

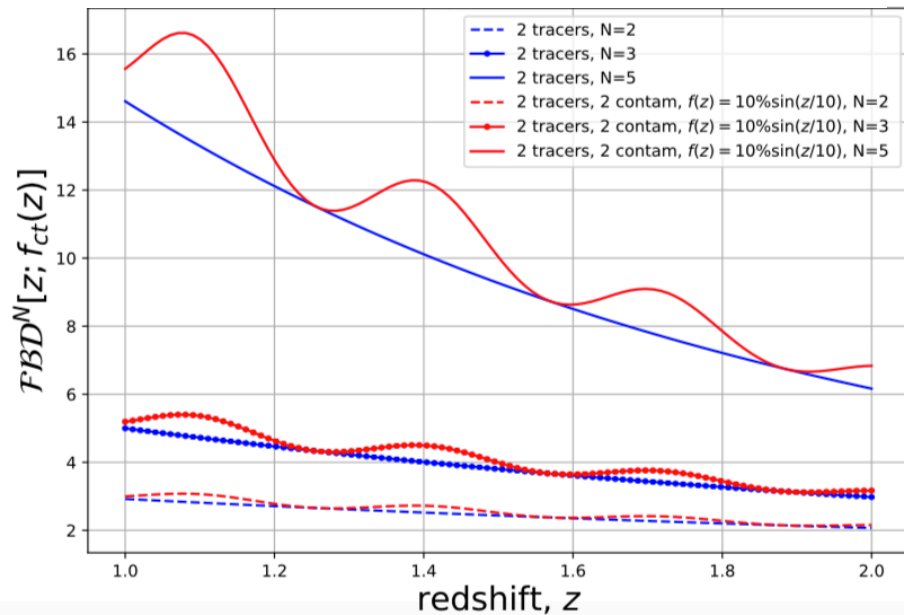
In this document, we introduce a novel formalism for any field theory and apply it to the effective field theories of large-scale structure. The new formalism is based on *functors of actions* composing those theories. This new formalism predicts the *actionic* fields. We discuss our findings in a *cosmological gravitology* framework. We present these results with a cosmological inference approach and give guidelines on how we can choose the best candidate between those models with some latest understanding of model selection using Bayesian inference.

A (D_τ, D_x) -manifold with NPCF of N_t -objects

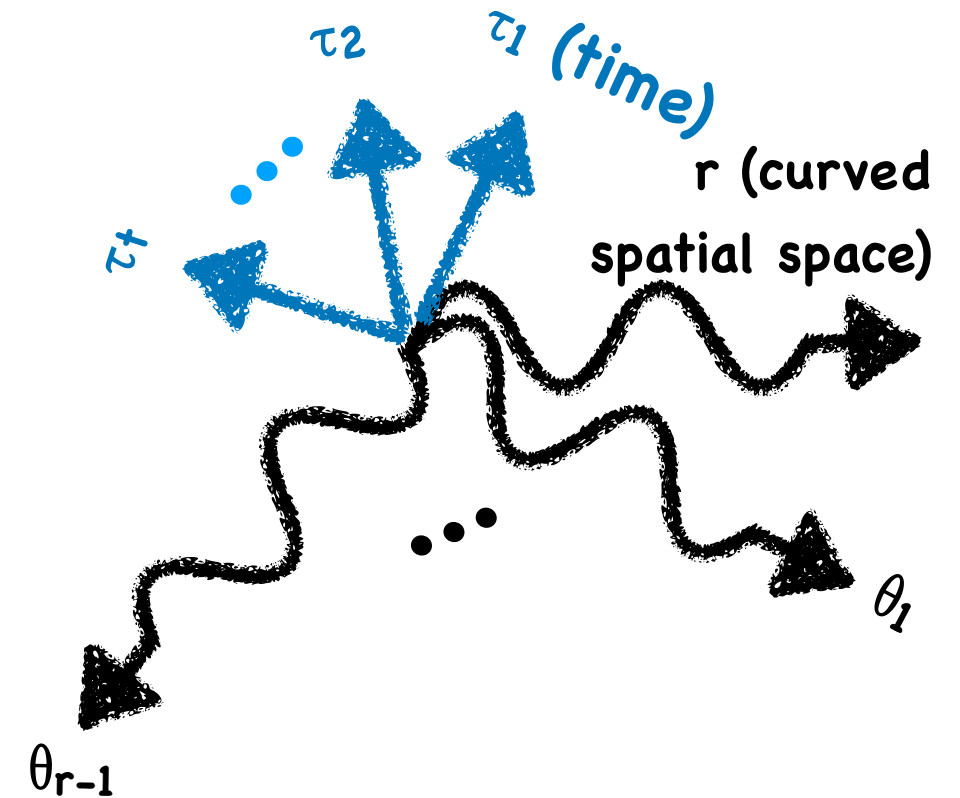
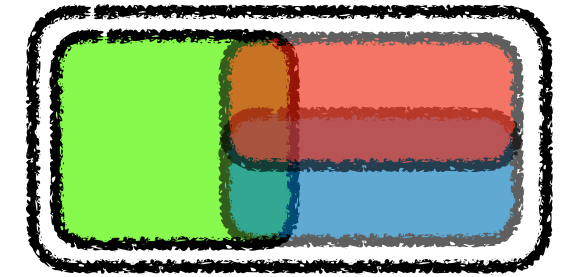
- extra dimensions
- contaminated targets
- Applied to Cosmic Structures:
 - Large scale structure (Astronomical)
 - Small scale structure (Quantum)

$$FBD(\vec{\tau}, \vec{x}) = \sum_{t=1}^{N_t^{OLSS}} \left\{ \left[1 - \sum_{c=1}^{N_{ct}} f_{ct}(\vec{\tau}, \vec{x}) \right] b_t(\vec{\tau}, \vec{x}) D_t(\vec{\tau}, \vec{x}) + \sum_{c=1}^{N_{ct}} |\vec{\gamma}_{ct}|^{D_x}(\vec{\tau}, \vec{x}) f_{ct}(\vec{\tau}, \vec{x}) b_{ct}(\vec{\tau}, \vec{x}) D_{ct}(\vec{\tau}, \vec{x}) \right\} \quad (3.19)$$

Functional Biased Growth

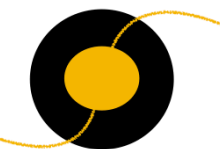
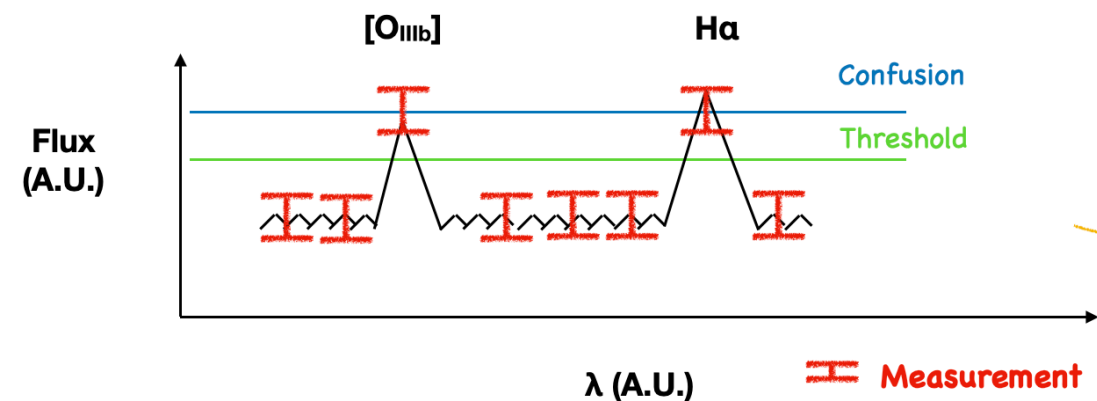


... Further Simulations ...



Suppose line-Identification on Flux

Line misidentification:
confusion of $H\alpha$ with $[O_{III}\lambda]$

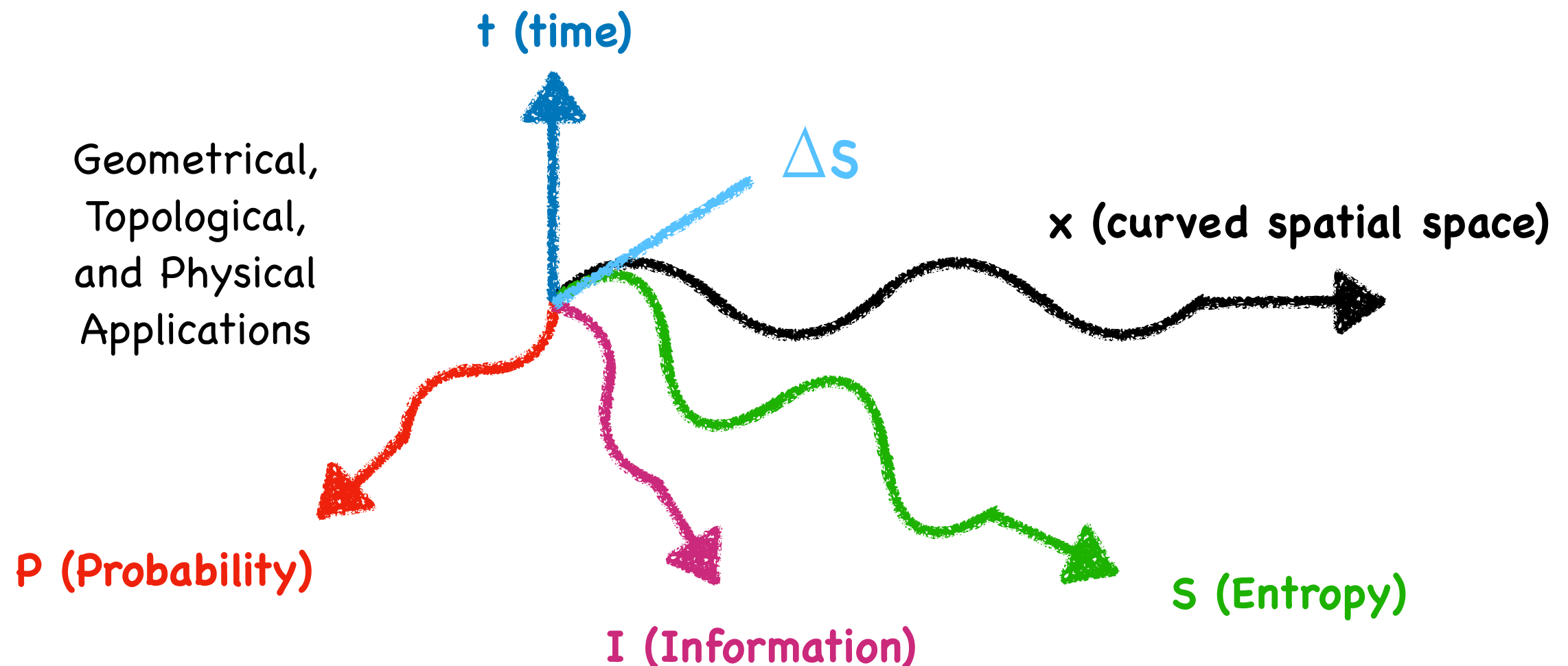


PN (2209.07472) IJGMMP Jan 2026

Advanced Manifold Metric Pairs

$$ds^{U+L} = \mathcal{F}_{\mu_t \mu_{\vec{x}}}^{\nu_{\vec{P}} \nu_{\vec{I}} \nu_{\vec{S}}} [c, f(c)] dc^{\mu_t} dc^{\mu_{\vec{x}}} dc_{\nu_{\vec{P}}} dc_{\nu_{\vec{I}}} dc_{\nu_{\vec{S}}}$$

$$c = [t, \vec{x}, \vec{P}, \vec{I}, \vec{S}]$$



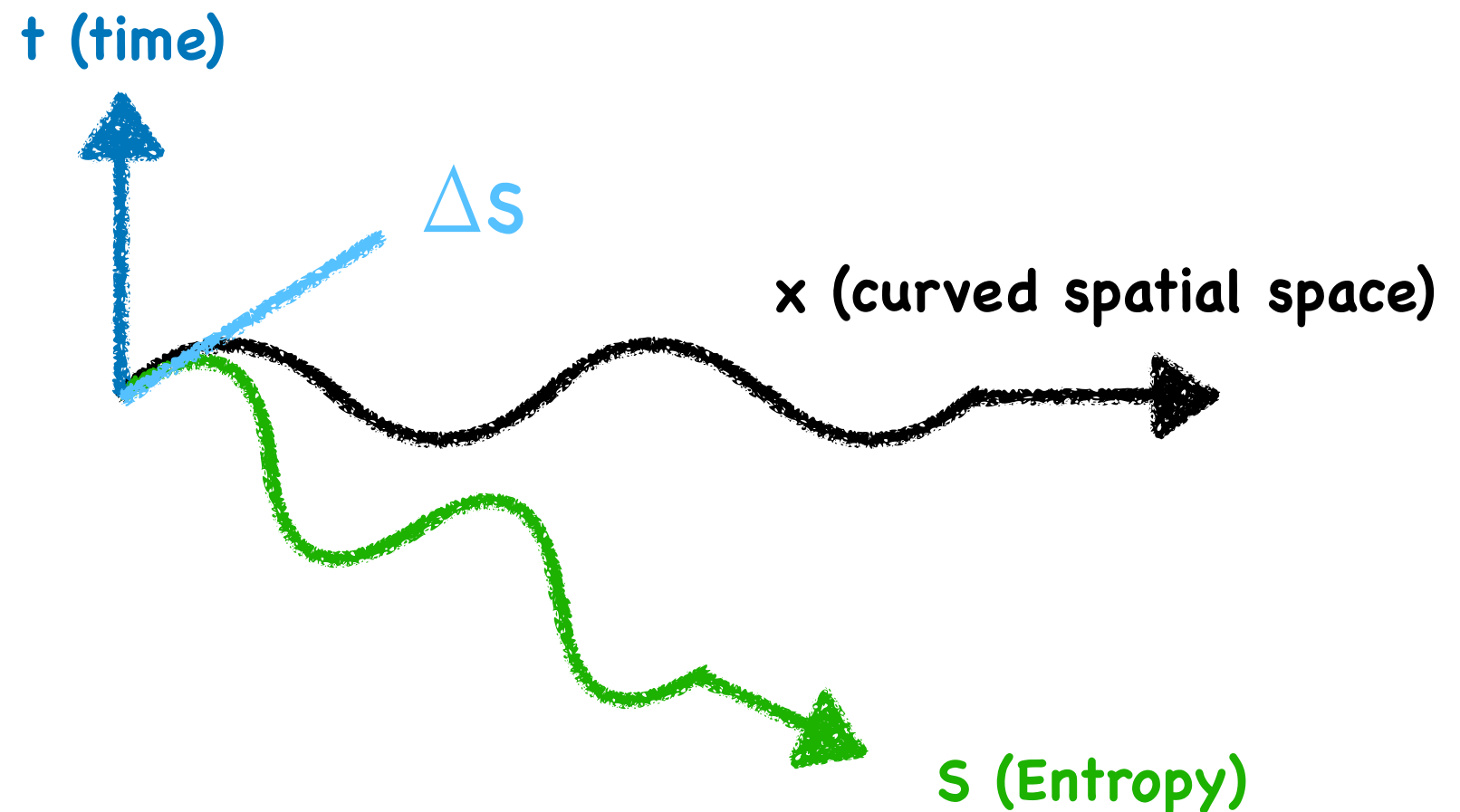
... Further Simulations ...

Published in MDPI Mathematics Journal
Visualisation via macos, keynote and zoom

Advanced Manifold Metric Pairs

$$ds^{U+L} = \mathcal{F}_{\mu_t \mu_{\vec{x}}}^{\nu_{\vec{P}} \nu_{\vec{I}} \nu_{\vec{S}}} [c, f(c)] dc^{\mu_t} dc^{\mu_{\vec{x}}} dc_{\nu_{\vec{P}}} dc_{\nu_{\vec{I}}} dc_{\nu_{\vec{S}}}$$

$$c = [t, \vec{x}, \cancel{\vec{P}}, \cancel{\vec{I}}, \vec{S}]$$



Published in MDPI Mathematics Journal
 Visualisation via macos, keynote and zoom

A temporal-spatial-entropic K-curvature

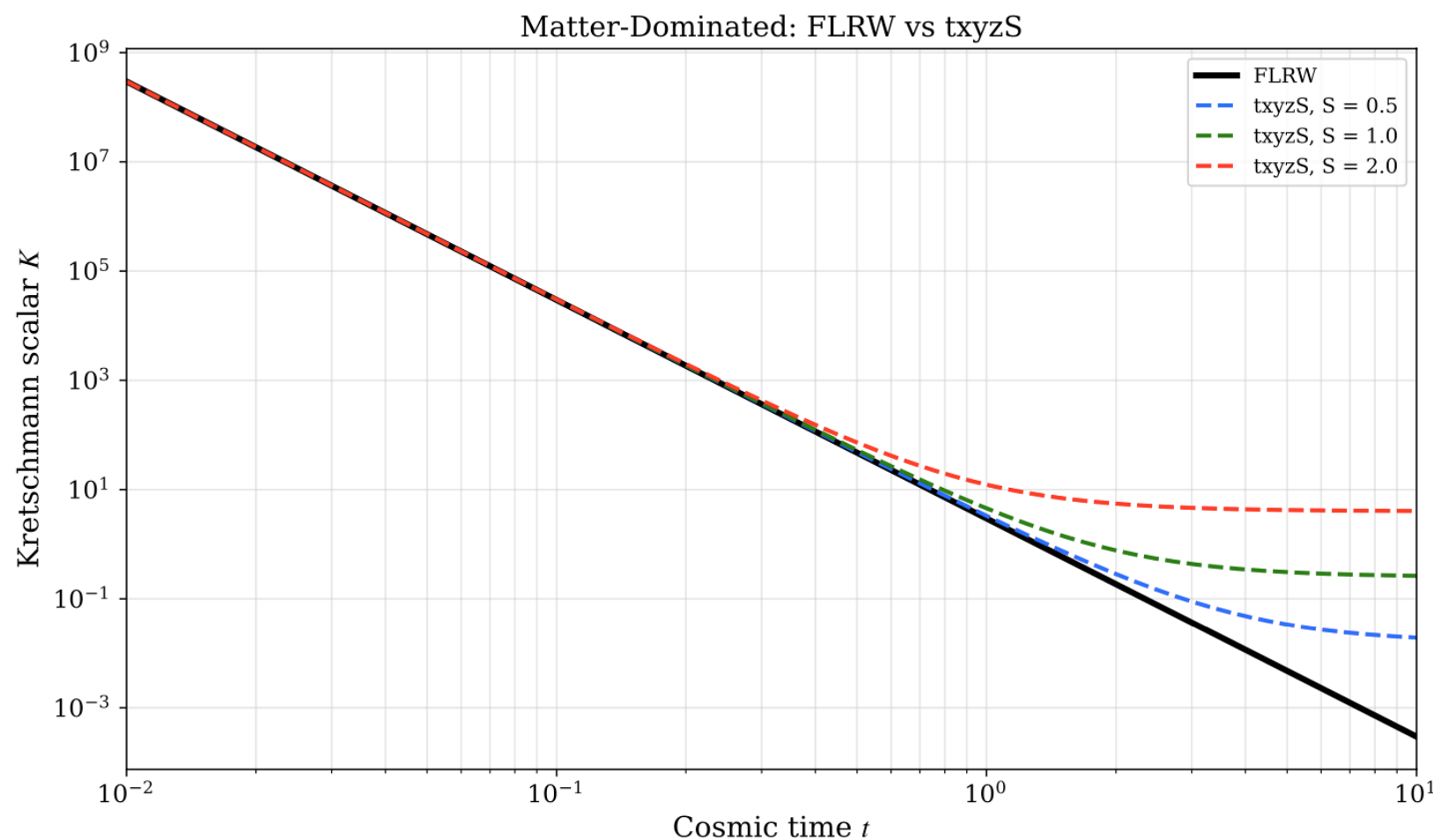
$$ds^2 = -c^2 dt^2 + a(t)^2(dx^2 + dy^2 + dz^2) + f_S(t, S)^2 dS^2$$

Considering its Kretschmann scalar

$$K = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta}$$

Compare txyzS to FLRW, with the ratio

$$\frac{K_{\text{txyzS}}}{K_{\text{FLRW}}} = 1 + \frac{\frac{4}{c^4}(\beta S)^4 + \frac{12}{c^4}\left(\frac{\dot{a}}{a}\right)^2(\beta S)^2}{\frac{12}{c^4}\left(\frac{\ddot{a}}{a}\right)^2 + \frac{12}{c^4}\left(\frac{\dot{a}}{a}\right)^4}$$



Predictions to be
constrained by

SnIa

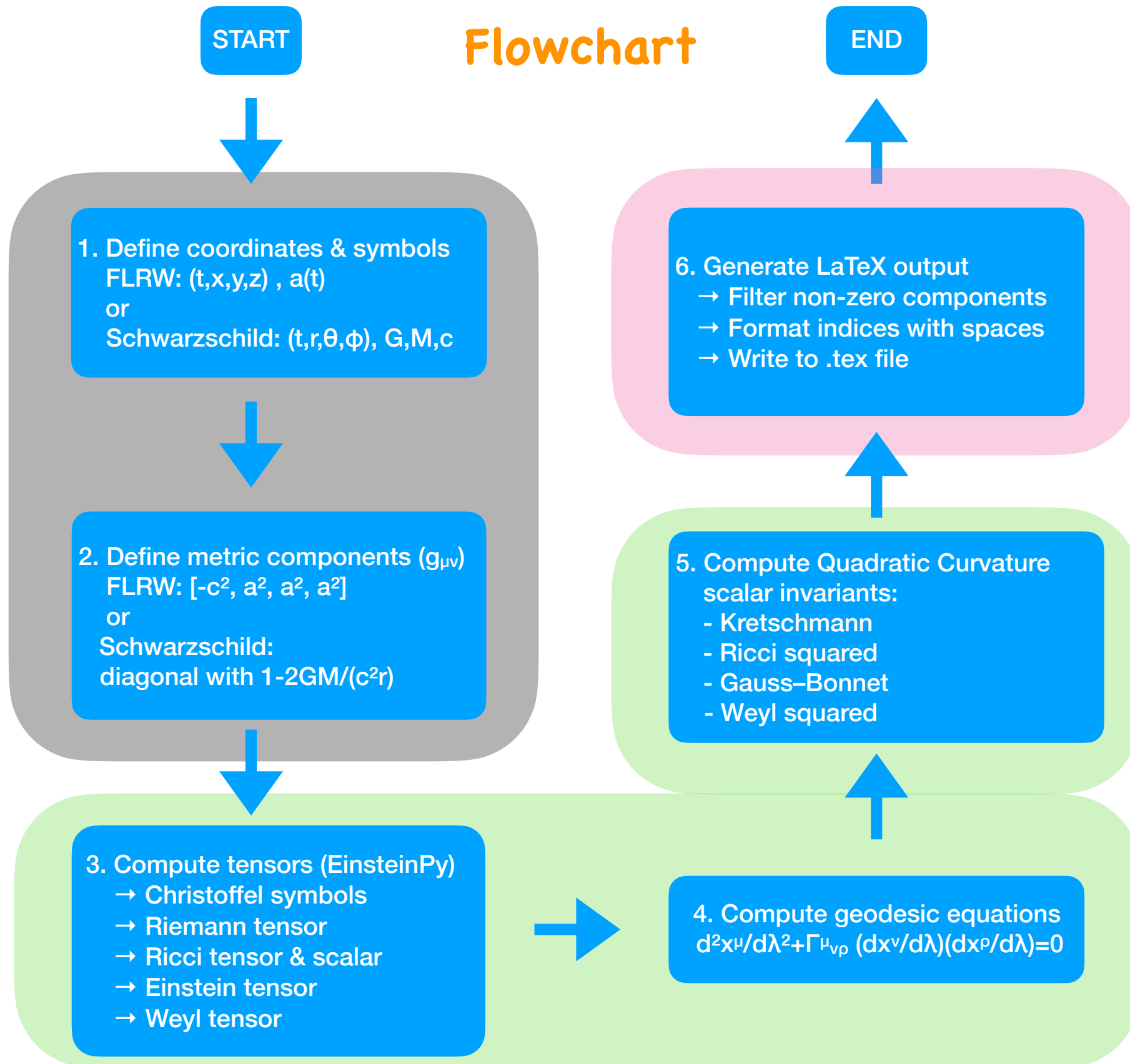
LSS

CMB

Figure 7: Comparison of Kretschmann scalar: FLRW vs txyzS (matter-dominated). The entropy dimension introduces an additional constant term $4(\beta S)^4$ and a coupling term $12H^2(\beta S)^2$, both increasing K relative to FLRW.

EinsteinKPy2Tex code

Flowchart



EinsteinKPy2Tex code

EinsteinKPy2Tex: A Python Tool for Computing and Exporting General Relativity Tensors to LaTeX

Pierros Ntelis^{†,‡}

[†]*School of Physics, Harbin Institute of Technology, Harbin 150001, People's Republic of China*

[‡]*Institute of Theoretical Physics, National University of Uzbekistan, Tashkent 100174, Uzbekistan*

June 7, 2026; conceived: 5 June 2026

Abstract

We present [EinsteinKPy2Tex](#), an open-source Python package that computes fundamental geometric quantities for the Friedmann-Lemaître-Robertson-Walker (FLRW) and Schwarzschild metrics using the EinsteinPy library. The code automatically calculates Christoffel symbols, Riemann curvature tensor, Ricci tensor, Ricci scalar, Einstein tensor, geodesic equations, and Kretschmann scalar. Results are exported as professionally formatted $\LaTeX$ documents, facilitating direct inclusion in research papers and educational materials.

The FLRW implementation is essential for cosmology, enabling the study of the universe's large-scale structure and dynamics. The Schwarzschild implementation is fundamental for black hole physics, describing the spacetime geometry around non-rotating compact objects. The code includes built-in citations for the original Einstein paper (1915) and the EinsteinPy library (Ribeiro et al. 2020), ensuring proper academic attribution.

[EinsteinKPy2Tex](#) is designed for educators, students, and researchers needing rapid, accurate tensor calculations with publication-ready output. The software is available at [GitHub](#) under an MIT license.

Keywords: General Relativity ; Schwarzschild metric ; FLRW metric ; Kretschmann scalar ; Geodesic equations ; Tensor calculus ; $\LaTeX$ export ; Symbolic computation ; EinsteinPy ; EinsteinKPy2Tex ; Cosmology ; Black hole physics

Contents

1	Introduction	2
2	Software Architecture	3
2.1	Dependencies	5
2.2	Code Structure	5
3	Physical Quantities Computed	5
3.1	Metric Tensor $g_{\mu\nu}$	5
3.2	Christoffel Symbols $\Gamma_{\nu\rho}^{\mu}$	5
3.3	Riemann Curvature Tensor $R_{\nu\rho\sigma}^{\mu}$	6
3.4	Ricci Tensor $R_{\mu\nu}$ and Scalar R	6
3.5	Einstein Tensor $G_{\mu\nu}$	6
3.6	Geodesic Equations	6
3.7	Kretschmann Scalar K	6

Easy to install

Efficient
documentation
and
examples

Flexible
to expand
and add
modules

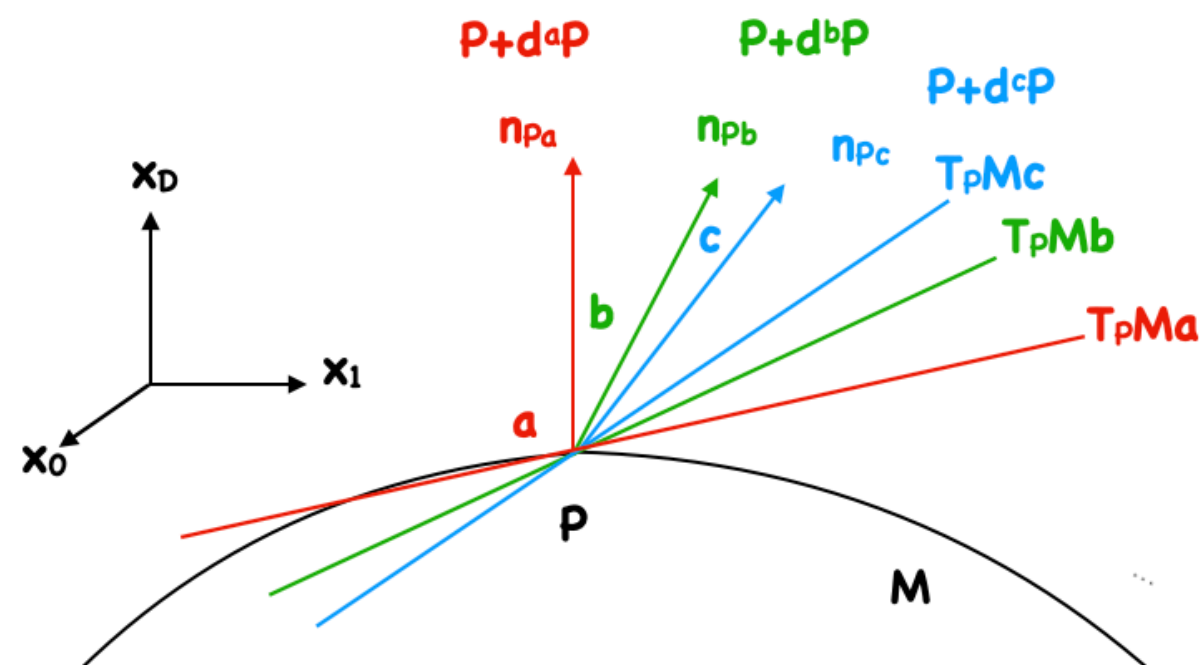
Appropriate
for
educational
and
research
purposes

Advancing Tensor Theories

the generic tensor, T , is simply the map between the generic combination of vectors, V , to the generic number set, G , written as

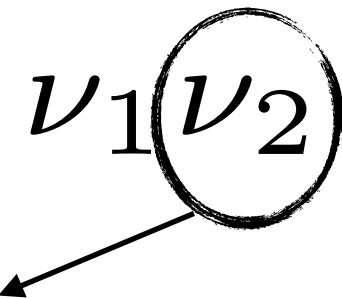
$$\mathcal{T} : \mathcal{V} \rightarrow \mathbb{G}$$

Geometrical and Topological interpretation is done using fractional calculus:



Advancing Tensor Theories

In the literature, there are tensor such as

$$T = T^{\mu_1 \mu_2}_{\nu_1 \nu_2}$$


Note that each index has its own index

=> NEW IDEA !

The index of the index ... of the index

Generic index

$$\mathcal{I} = \ell_{k \dots j_i}$$

=> Generic tensor !

$$T = T^{\ell_{k \dots j_i}}_{\alpha_{\beta \dots \psi_{\omega}}}$$

New mathematical object!

A Foundational Analysis of Kernel-based Calculus (AFAKEC)

Kernel Based Derivative

$$D_x^{F,\alpha} f(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x F(x, \alpha)) - f(x)}{\Delta x}$$

$$D_x^{F,\alpha} f(x) = F(x, \alpha) f'(x)$$

Kernel Based Chain Rule

$$D_x^{F,\alpha} [f(g(x))] = D_g^{F,\alpha} [f(g)] \cdot D_x^{F,\alpha} [g(x)] \cdot F^{-1}(g(x), \alpha)$$

Kernel Based Integral

$$I_x^{F,\alpha} f(x) = \int d_{F,\alpha} x f(x) = \int \frac{dx}{F(x, \alpha)} f(x)$$

Fundamental Theorem of Local Kernel-based Calculus (FTLKEC)

$$\int_{x_{\min}}^{x_{\max}} \frac{D_x^{F,\alpha} f(x)}{F(x, \alpha)} dx = f(x_{\max}) - f(x_{\min})$$

Proof:

$$\int_{x_{\min}}^{x_{\max}} \frac{D_x^{F,\alpha} f(x)}{F(x, \alpha)} dx = \int_{x_{\min}}^{x_{\max}} \frac{F(x, \alpha) D_x f(x)}{F(x, \alpha)} dx = \int_{x_{\min}}^{x_{\max}} f'(x) dx = f(x_{\max}) - f(x_{\min})$$

A Foundational Analysis of Kernel-based Calculus (AFAKEC)

Identity operator of Local Kernel-based Calculus

$$I_t^{F,\alpha,x_0,x} \left(D_t^{F,\alpha} f(t) \right) = f(x) - f(x_0)$$

Proof:

$$I_t^{F,\alpha,x_0,x} \left(D_t^{F,\alpha} f(t) \right) = \int_{x_0}^x \frac{D_t^{F,\alpha} f(t)}{F(t,\alpha)} dt = \int_{x_0}^x \frac{f'(t) F(t,\alpha)}{F(t,\alpha)} dt = \int_{x_0}^x f'(t) dt = f(x) - f(x_0)$$

Remark 21. Conversely, differentiating a local kernel-based integral returns the original function:

$$D_x^{F,\alpha} \left(I_x^{F,\alpha} f(x) \right) = D_x^{F,\alpha} \left(\int \frac{f(x)}{F(x,\alpha)} dx \right) = \frac{d}{dx} \left(\int \frac{f(x)}{F(x,\alpha)} dx \right) \cdot F(x,\alpha) \quad (142)$$

$$= \frac{f(x)}{F(x,\alpha)} \cdot F(x,\alpha) = f(x). \quad (143)$$

Thus $D_x^{F,\alpha} I_x^{F,\alpha} = I$ on the space of differentiable functions where the relevant integrals exist and $F(x,\alpha) \neq 0$. For more general function classes (e.g., merely integrable functions), the identity holds only in the almost-everywhere sense or under additional regularity assumptions.

Fractional Calculus in Cosmology

with the N. Uzbekistan University Group

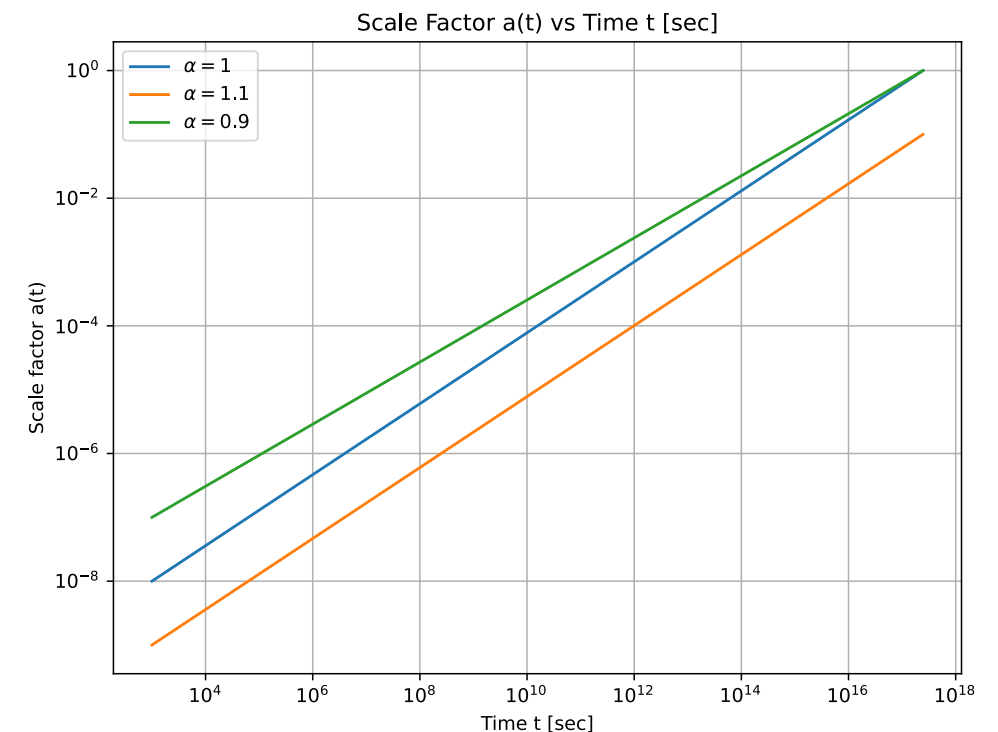
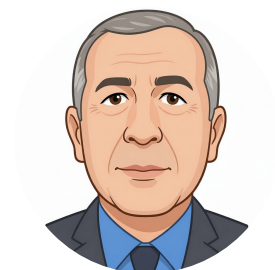
– Cosmic applications with
Pakhlavon Yovkochev



– Foundational aspects with
Bakhodir Shodikulov



Supervised by B.J. Ahmedov, and PN



Fiber bundle:
base space B ,
total space E ,
fiber F ,
projections π .

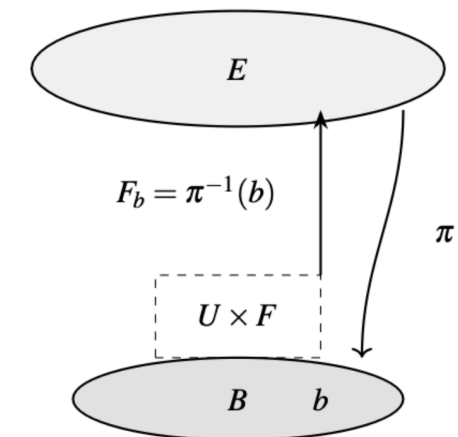
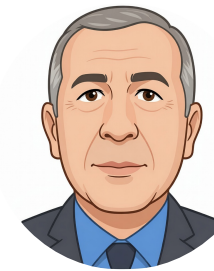
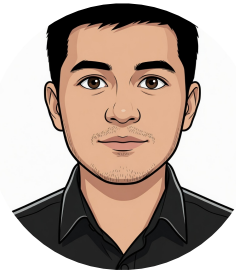


Figure 1: Schematic of a Fibre bundle (E, B, π, F) .

Preliminary results

Chaotic motion and power spectral density in Schwarzschild Bertotti–Robinson black hole spacetime

by Y. Xu, U. Uktamov, PN, A. Abdujabbarov, B.J. Ahmedov, C. Yuan



$$ds^2 = \frac{1}{R^2} \left[-Q dt^2 + \frac{dr^2}{Q} + r^2 \left(\frac{d\theta^2}{P} + P \sin^2 \theta d\phi^2 \right) \right]$$

$$P = 1 + B^2 M^2 \cos^2 \theta$$

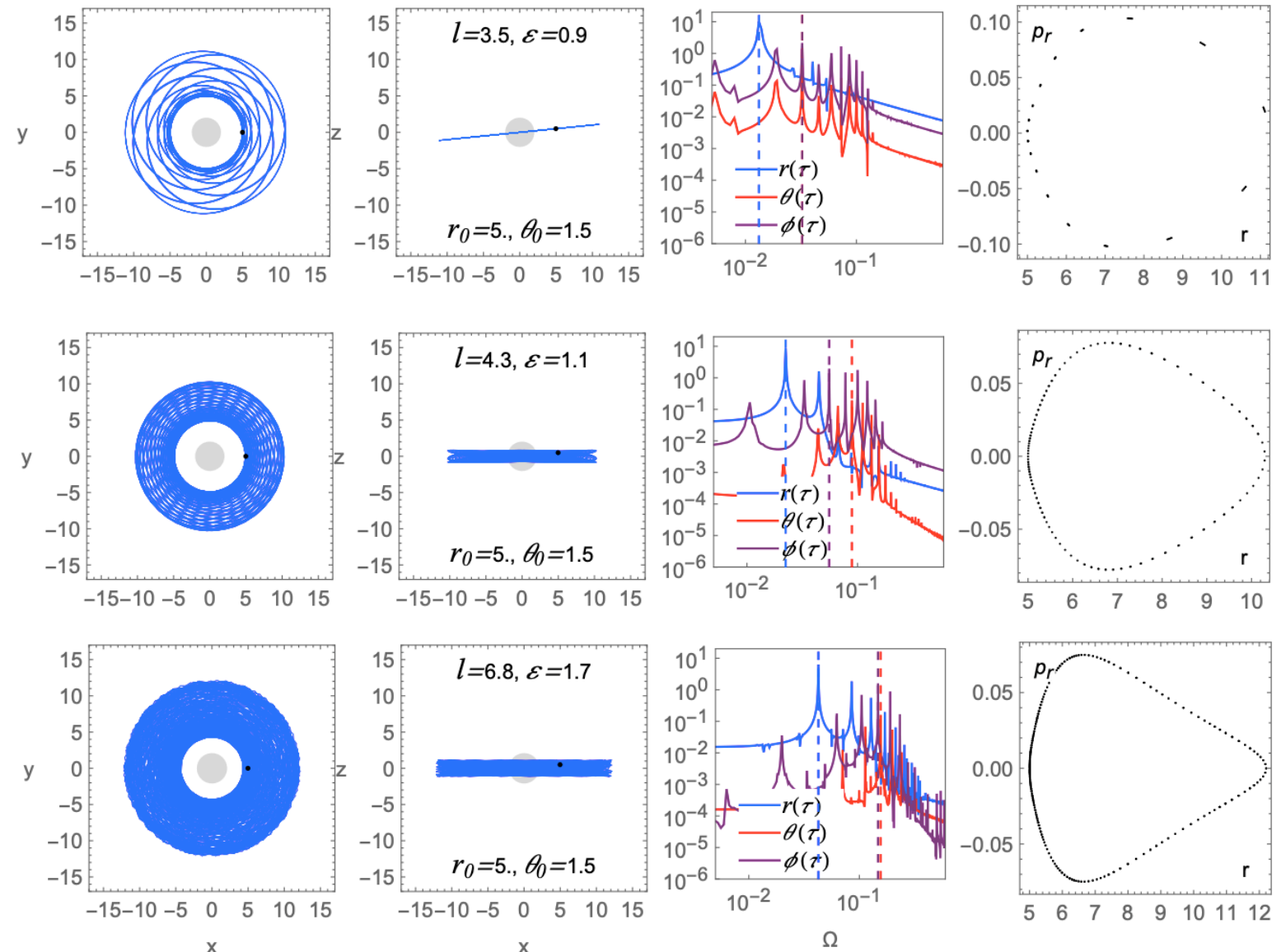
$$Q = (1 + B^2 r^2) \left(1 - B^2 M^2 - \frac{2M}{r} \right)$$

$$R^2 = 1 + B^2 [r^2 \sin^2 \theta + (2Mr + B^2 M^2 r^2) \cos^2 \theta]$$

M = mass field of BH

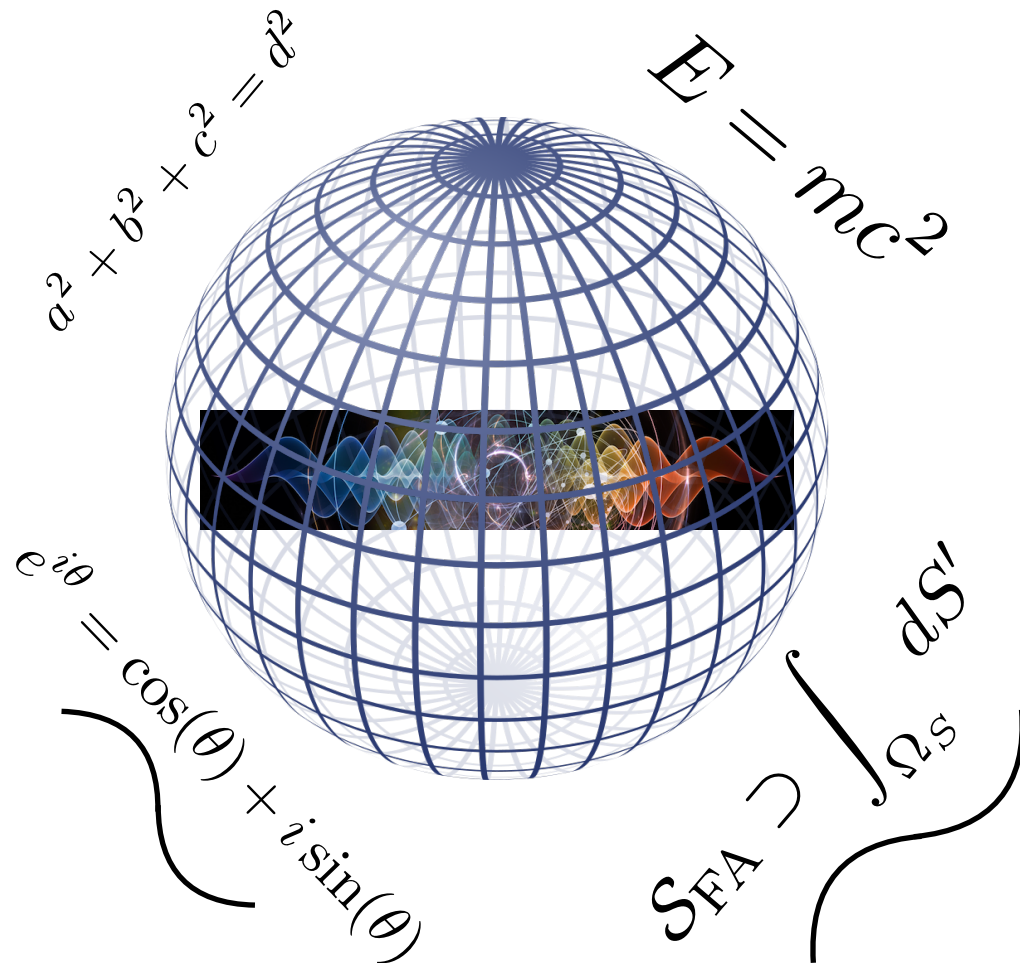
B = Magnetic Field around BH
distorting the spacetime

Charged Test particle
stabilises its orbit,
as Magnetic Field increases



Science Popularisation !

Universions



YouTube



Analogue



Thank you

谢谢

Xiè xiè

! :)



Subscribe to support Art & Science ! :)

Conclusions and Outlook

- Mathematical and Observational evidence for
 - Functors of actions theories and predictions of actionions
 - Quintessence ($\varphi\Lambda\text{CDM}$)
- Dynamical analysis on
 - Standard cosmology (ΛCDM)
 - Quintessence ($\varphi\Lambda\text{CDM}$) (Complex and Simple DAC)
 - $\text{poly}\Lambda\text{CDM}$ model
- Advancing Manifold-Metric Pairs
 - Functional (U,L)-rank tensor metric
 - Restructuring extra dimensions with t, x, P, I, S
- A (D_τ, D_x) -manifold with NPCF of N_t -objects
- Advancing Tensor Theories
 - Generic Indices, Generic Tensors
 - Generic Tensors connected with Fractional Calculus
 - Fractional Calculus in Cosmology

Thank you

谢谢

Xiè xiè

Ευχαριστώ

! :)